Small-cell lung cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up

SUPPLEMENTARY MATERIAL

SECTION 1 – DIAGNOSIS AND PATHOLOGY/MOLECULAR BIOLOGY

Diagnosis and pathology

Small-cell lung cancer (SCLC) is defined by the 2015 World Health Organization (WHO) classification as a high-grade neuroendocrine tumour composed of tumour cells measuring less than three resting lymphocytes, with scant cytoplasm, poorly defined cell borders, a finely dispersed granular nuclear chromatin and absent or inconspicuous nucleoli. Mitoses are numerous, with an average of 80 mitoses per 2 mm^2 area, and necrosis is extensive.[1] On small biopsies, crush artifacts can be predominant. On sputum and bronchial washings, tumour cells appear larger but with cytoplasm restricted to small rims. Combined SCLC is defined by a mixture of pure SCLC areas and of adenocarcinoma, squamous cell carcinoma (SCC), large cell carcinoma or sarcomatoid (spindle or giant cell) carcinoma areas, regardless of the amount of non-small cell lung cancer (NSCLC) component.[1] In contrast, for combined large-cell neuroendocrine carcinoma (LCNEC) and SCLC, a minimum of 10% LCNEC is required. Combined SCLC shares a similar epidemiology and clinical presentation as pure SCLC, although combined SCLC primary tumours tend to be located more peripherally in the lung. Transformation to SCLC is a known resistance mechanism in epidermal growth factor receptor (EGFR)-mutated adenocarcinoma of the lung treated with EGFR tyrosine kinase inhibitors.[2] It occurs in 3%-5% of patients and is associated with loss of *RB1* and *TP53*.[3]

Immunohistochemistry for neuroendocrine markers can be of great help, particularly on biopsies. The majority of SCLC diffusely expresses CD56 (membranous staining), whereas synaptophysin and chromogranin A cytoplasmic staining are observed in 54% and in 37% of the cases, respectively.[4] CD56 expression is considered the most sensitive but the least specific marker.[5] Of note, 10%-25% of SCLC can be negative or focally positive for only one marker.[6,7] Thyroid transcription factor 1 (TTF1) is expressed by 90% of SCLC and >50% of nuclei stain

positively for Ki-67.[8-11] An International Association for the Study of Lung Cancer (IASLC) pathology committee study has shown that the agreement rate between pathologists increased from 0.55 to 0.60 when these markers were used.[5] In addition, SCLC frequently expresses cytokeratins, often with a 'punctate and paranuclear dot' pattern[5,12,13] and is negative for P40, a typical marker of SCC, with a cut-off of 10% positive cells.[14,15] Differential diagnosis includes other neuroendocrine lung tumours, such as carcinoid tumours and LCNEC, other NSCLC with small basal-like cells (basaloid carcinoma, NUT1 carcinoma) and more rarely, small round cell sarcoma, metastatic breast carcinoma and non-Hodgkin lymphoma.

There has been recent interest in trying to define predictive biomarkers for immunotherapy in SCLC. Both the prevalence and strength of programmed death-ligand 1 (PD-L1) expression in SCLC tumour cells appear substantially lower than that seen in other solid tumours, including NSCLC.[16] PD-L1 expression by tumour cells was reported in less than 20% of SCLC with the 28.8 clone[16,17] or the SP42 clone[16,18] and in 2%-10% with the 22C3 clone[19,20] using a 1% tumour proportion score (TPS) threshold. Less than 5% exhibit a TPS >50% with the 22C3 clone.[21] In contrast, stroma immune cells express PD-L1 in up to 40% of cases using 22C3 antibody at the same cut-off.[20,21] While more studies are needed, available data reported in the IMpower133, CASPIAN and KEYNOTE-158 and -028 trials do not support PD-L1 staining assessment as a predictive biomarker for immunotherapy efficacy. Of interest, in the two latter studies, 14 of 16 responders were PD-L1-positive using PD-L1 combined positive score (CPS) >1, defined as the number of PD-L1 staining cells (tumour cells, lymphocytes, macrophages) divided by the total number of viable tumour cells. A high tumour mutational burden (TMB) was reported in SCLC with a range of 5.5-8.62 non synonymous mutations per megabase.[22,23] In the Checkmate 032 trial, the overall response rate was higher for patients receiving nivolumab or nivolumab plus ipilimumab (21.3% and 46.2%, respectively) and exhibiting a high TMB, defined as ≥248 somatic missense mutations (highest tertile) detected by whole exome sequencing.[24] There was no correlation between TMB and PD-L1 expression. In the IMpower 133 trial, using a combination of atezolizumab and chemotherapy (ChT), reducing the biomarker discriminative capacity for immunotherapy activity and blood TMB (bTMB) using cut-offs of 10 and 16 mutations per megabase on the Foundation Medicine platform, was not predictive for

outcome.[25] The predictive value of TMB for immunotherapy requires prospective confirmation and is not currently considered for routine practice treatment decision-making.

Molecular biology

Studies of genome-wide analyses of mutations, translocations and copy number alterations conducted in North America, Europe and Asia have all confirmed characteristic loss-of-function mutations of both *TP53* and *RB1* in almost all SCLCs.[23,26-28] They have identified 'mutually exclusive' mutational inactivation of either of two histone acetyltransferases, EP300 and CREBBP, and a recent analysis in genetically engineered mouse models suggests that CREBBP inactivation may sensitise SCLC tumours to histone deacetylase inhibitors.[29] Frequent inactivation of Notch family members was also observed.[23,29] Inactivation of Notch signalling is involved in developmental differentiation of neuroendocrine tissues, and mouse modelling of oncogenesis demonstrates that enforced Notch signalling can effectively suppress the development of SCLC.[23] The commonly recurring mutations in SCLC, including *TP53*, *RB1*, multiple epigenetic regulators and Notch family members, are all loss-of-function, limiting the ability to directly target oncogenic mutations therapeutically. SCLC in never smokers is rare (<2%) and is related to a lower frequency of *TP53* mutations and *RB1* loss.[30] Some cases of *de novo* EGFR-mutated SCLC have been described; however, response to EGFR-directed therapy appears to be poor.[30]

Recent epigenetic and global gene expression profiling has suggested biologically distinct SCLC subtypes, defined by key transcriptional regulators. Two transcription factors previously implicated as critical determinants of neuroendocrine fate decisions in normal development, ASCL1 and NEUROD1, are differentially expressed, and data from both human and mouse tumours suggest that these are important oncogenic factors, particularly ASCL1 which is essential for SCLC tumourigenesis in a mouse model with inactivation of *TP53* and *RB1.* [31-34] Other recent analyses of human cell lines and tumours suggest two other smaller subtypes of SCLC, one highly dependent on a third critical transcriptional regulator, POU2F3, and another notable for high expression of the transcriptional regulator, YAP1.[35,36] A

recent consensus review from multiple laboratory investigators proposed a unified nomenclature for these subtypes: SCLC-A, -N, -P and -Y, for subtypes with dominant expression of ASCL1, NEUROD1, POU2F3 and YAP1, respectively.[37]

SECTION 2 – STAGING OF SCLC

Supplementary Table S1. Clinical classification of SCLC according to the UICC TNM 8th edition

T – primary tumour	
TX	Primary tumour cannot be assessed or tumour proven by the presence of malignant cells in sputum or bronchial washings but not visualised by imaging or bronchoscopy
T0	No evidence of primary tumour
Tis	Carcinoma *in situ*[a]
T1	Tumour ≤3 cm in the greatest dimension surrounded by lung or visceral pleura and without bronchoscopic evidence of invasion more proximal than the lobar bronchus (i.e. not in the main bronchus)[b] T1mi Minimally invasive adenocarcinoma[c] T1a Tumour ≤1 cm in the greatest dimension[b] T1b Tumour >1 cm but ≤2 cm in the greatest dimension[b] T1c Tumour >2 cm but ≤3 cm in the greatest dimension[b]

T2	Tumour >3 cm but ≤5 cm, or a tumour with any of the following features:[d] ● Involves main bronchus regardless of distance to the carina but without involvement of the carina ● Invades visceral pleura ● Associated with atelectasis or obstructive pneumonitis that extends to the hilar region either involving part of or the entire lung T2a Tumour >3 cm but ≤4 cm in the greatest dimension T2b Tumour >4 cm but ≤5 cm in the greatest dimension
T3	Tumour >5 cm but ≤7 cm in the greatest dimension or one that directly invades any of the following: parietal pleura, chest wall (including superior sulcus tumours) phrenic nerve, parietal pericardium or separate tumour nodule(s) in the same lobe as the primary
T4	Tumour >7 cm or of any size that invades any of the following: diaphragm, mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, oesophagus, vertebral body, carina or separate tumour nodule(s) in a different ipsilateral lobe to that of the primary
N – regional lymph nodes	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in ipsilateral peri-bronchial and/or ipsilateral hilar lymph nodes and intrapulmonary nodes, including involvement by direct extension

N2	Metastasis in ipsilateral mediastinal and/or subcarinal lymph node(s)
N3	Metastasis in contralateral mediastinal, contralateral hilar, ipsilateral or contralateral scalene or supraclavicular lymph node(s)
M – distant metastasis	
M0	No distant metastasis
M1	Distant metastasis M1a Separate tumour nodule(s) in a contralateral lobe; tumour with pleural or pericardial nodules or malignant pleural or pericardial effusion[e] M1b Single extrathoracic metastasis in a single organ[f] M1c Multiple extrathoracic metastasis in a single organ or multiple organs

SCLC, small-cell lung cancer; TNM, tumour, node, metastasis; UICC, Union for International Cancer Control.

[a] Tis includes adenocarcinoma *in situ* and squamous carcinoma *in situ*.

[b] The uncommon superficial spreading tumour of any size with its invasive component limited to the bronchial wall, which may extend proximally to the main bronchus, is also classified as T1a.

[c] Solitary adenocarcinoma (≤3 cm in the greatest dimension) with a predominantly lepidic pattern and ≤5 mm invasion in the greatest dimension in any one focus.

[d] T2 tumours with these features are classified T2a if ≤4 cm or if size cannot be determined, and T2b if >4 cm but ≤5 cm.

[e] Most pleural (pericardial) effusions with lung cancer are due to the tumour. In a few patients, however, multiple microscopic examinations of pleural (pericardial) fluid are negative for tumour, the fluid is non-bloody and is not an exudate. Where these

elements and clinical judgment dictate that the effusion is not tumour-related, the effusion should be excluded as a staging descriptor.

[f] This includes involvement of a single non-regional node.

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Supplementary Table S2. Staging and stage grouping of SCLC according to the UICC TNM 8th edition

Stage	T	N	M
Occult carcinoma	X	0	0
0	Is	0	0
IA	1	0	0
IA1	1mi 1a	0	0
IA2	1b	0	0
IA3	1c	0	0
IB	2a	0	0
IIA	2b	0	0
IIB	1a-c, 2a,b 3	1 0	0 0
IIIA	1a-c, 2a,b 3 4	2 1 0, 1	0 0 0
IIIB	1a-c, 2a,b 3, 4	3 2	0 0
IIIC	3, 4	3	0
IV	Any	Any	1
IVA	Any	Any	1a, 1b
IVB	Any	Any	1c

SCLC, small-cell lung cancer; TNM, tumour, node, metastasis; UICC, Union for International Cancer Control.

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SECTION 3 – ESMO-MCBS

Supplementary Table S3. ESMO-MCBS table for new therapies/indications in SCLC

Therapy	Disease setting	Trial	Control	Absolute survival gain	HR (95% CI)	QoL/ toxicity	ESMO-MCBS score[a]
Atezolizumab + carboplatin and etoposide	First-line treatment of extensive-stage SCLC	A study of carboplatin plus etoposide with or without atezolizumab in participants with untreated extensive-stage SCLC (IMpower133)[25,39,40] Phase III NCT02763579	Placebo + carboplatin and etoposide Median PFS: 4.3 months Median OS: 10.3 months	PFS gain: 0.9 months OS gain: 2.0 months	PFS HR: 0.77 (0.62-0.96) OS HR: 0.76 (0.60-0.95)	Benefit for delayed deterioration in global QoL not significant	3 (Form 2a)
Durvalumab + etoposide	First-line treatment of	Durvalumab ± tremelimumab in	Platinum + etoposide			Benefit for delayed	3

and cisplatin or carboplatin	extensive-stage SCLC	combination with platinum-based ChT in untreated extensive-stage SCLC (CASPIAN)[41-43] Phase III NCT03043872	Median OS: 10.5 months	OS gain: 2.4 months	OS HR: 0.75 (0.62-0.91)	deterioration in global QoL not significant	(Form 2a)
Lurbinectedin	Metastatic (SCLC) with disease progression on or after platinum-based ChT	Lurbinectedin as second-line treatment for patients with small-cell lung cancer (PM1183-B-005-14)[44] Phase II NCT02454972	Single arm (no control)		ORR: 35.2% DoR: 5.3 months PFS: 3.5 months		1 (Form 3)

ChT, chemotherapy; CI, confidence interval; DoR; duration of response, ESMO-MCBS, European Society for Medical Oncology-Magnitude of Clinical Benefit Scale; HR, hazard ratio; ORR; overall response rate, OS, overall survival; PFS, progression-free survival; QoL, quality of life, SCLC, small-cell lung cancer.

[a] ESMO-MCBS version 1.1.[45] The scores have been calculated by the ESMO-MCBS Working Group and validated by the ESMO Guidelines Committee (https://www.esmo.org/guidelines/esmo-mcbs/scale-evaluation-forms-v1.0-v1.1/scale-evaluation-forms-v1.1).

SECTION 4 – LEVELS OF EVIDENCE AND GRADES OF RECOMMENDATION

Supplementary Table S4. Levels of evidence and grades of recommendation (adapted from the Infectious Diseases Society of America-United States Public Health Service Grading System[a])

Levels of evidence

I	Evidence from at least one large randomised, controlled trial of good methodological quality (low potential for bias) or meta-analyses of well-conducted randomised trials without heterogeneity
II	Small randomised trials or large randomised trials with a suspicion of bias (lower methodological quality) or meta-analyses of such trials or of trials demonstrated heterogeneity
III	Prospective cohort studies
IV	Retrospective cohort studies or case-control studies
V	Studies without control group, case reports, expert opinions

Grades of recommendation

A	Strong evidence for efficacy with a substantial clinical benefit, strongly recommended
B	Strong or moderate evidence for efficacy but with a limited clinical benefit, generally recommended
C	Insufficient evidence for efficacy or benefit does not outweigh the risk or the disadvantages (adverse events, costs, etc.), optional

D	Moderate evidence against efficacy or for adverse outcome, generally not recommended
E	Strong evidence against efficacy or for adverse outcome, never recommended

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